

HW1 (Chapters 2 & 3) for MSE 201 Spring 2023

Textbook:

W. D. Callister and D.G. Rethwisch, Materials Science and Engineering: An Introduction, 8th Edition

Chapter 2

Problem 2.14

2.14 The net potential energy between two adjacent ions, E_N , may be represented by the sum

of Equations 2.8 and 2.9; that is,

$$E_N = -\frac{A}{r} + \frac{B}{r^n} \quad (2.11)$$

Calculate the bonding energy E_0 in terms of the parameters A , B , and n using the following procedure:

1. Differentiate E_N with respect to r , and then set the resulting expression equal to zero, because the curve of E_N versus r is a minimum at E_0 .
 2. Solve for r in terms of A , B , and n , which yields r_0 , the equilibrium interionic spacing.
 3. Determine the expression for E_0 by substituting r_0 into Equation 2.11.
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Problem 2.15

2.15 For a $\text{K}^+ - \text{Cl}^-$ ion pair, attractive and repulsive energies E_A and E_R , respectively, depend on the distance between the ions r , according to

$$E_A = -\frac{1.436}{r}$$
$$E_R = \frac{5.86 \times 10^{-6}}{r^9}$$

For these expressions, energies are expressed in electron volts per $\text{K}^+ - \text{Cl}^-$ pair, and r is the distance in nanometers. The net energy E_N is just the sum of the preceding two expressions.

- (a) Superimpose on a single plot E_N , E_R , and E_A versus r up to 1.0 nm.
 - (b) On the basis of this plot, determine (i) the equilibrium spacing r_0 between the K^+ and Cl^- ions, and (ii) the magnitude of the bonding energy E_0 between the two ions.
 - (c) Mathematically determine the r_0 and E_0 values using the solutions to Problem 2.14 and compare these with the graphical results from part (b).
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Problem 2.20

2.20 Make a plot of bonding energy versus melting temperature for the metals listed in Table 2.3. Using this plot, approximate the bonding energy for copper, which has a melting temperature of 1084°C .

Chapter 3

Problem 3.7

3.7 Iron has a BCC crystal structure, an atomic radius of 0.124 nm, and an atomic weight of 55.85 g/mol. Compute and compare its theoretical density with the experimental value found inside the front cover.

Problem 3.17

3.17 Titanium has an HCP unit cell for which the ratio of the lattice parameters c/a is 1.58. If the radius of the Ti atom is 0.1445 nm, **(a)** determine the unit cell volume, and **(b)** calculate the density of Ti and compare it with the literature value.
